

Lecture 11: The First Law of Thermodynamics – Tutorial Sheet

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Use the Master Procedure: identify the system and the substance, sketch the path on a pV diagram, fix the signs (Q into the system, W done by the system), predict, select the model, execute with an energy ledger, evaluate. Problems A1 and A2 are solved in full; final answers to the others are at the end. Useful values: $R = 8.314 \text{ J/(mol K)}$, $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$, $1 \text{ kcal} = 4186 \text{ J}$, $g = 9.8 \text{ m/s}^2$. Monatomic gas: $C_V = \frac{3}{2}R$, $C_p = \frac{5}{2}R$, $\gamma = \frac{5}{3}$. Diatomic gas (room temperature): $C_V = \frac{5}{2}R = 20.79$, $C_p = \frac{7}{2}R = 29.10 \text{ J/(mol K)}$, $\gamma = 1.40$. Water: $L_v = 2.256 \times 10^6 \text{ J/kg}$, $L_f = 3.34 \times 10^5 \text{ J/kg}$; 1.00 g of liquid water at 100°C becomes 1671 cm^3 of steam at 1 atm . Treat all gases as ideal and all processes as quasi-static, unless told otherwise.

A. Worked examples

A1. A three-leg cycle of argon

0.60 mol of argon (monatomic) is at $2.0 \times 10^5 \text{ Pa}$ and 300 K (state 1). It is heated at constant volume until its pressure is doubled (state 2), then expands isothermally until its pressure is again $2.0 \times 10^5 \text{ Pa}$ (state 3), then is compressed at constant pressure back to state 1. Find Q , W and ΔU for each leg, and the net work and net heat.

Solution. *Sketch:* a vertical line up, a falling hyperbola to the right, and a horizontal line back to the left: the path runs clockwise, so the net work should be positive. *States:* $V_1 = \frac{nRT_1}{p_1} = \frac{(0.60)(8.314)(300)}{2.0 \times 10^5} = 7.48 \times 10^{-3} \text{ m}^3$ (7.48 L). *State 2:* $V_2 = V_1$, $T_2 = 600 \text{ K}$, $p_2 = 4.0 \times 10^5 \text{ Pa}$. *State 3:* $T_3 = 600 \text{ K}$, $p_3 = 2.0 \times 10^5 \text{ Pa}$, so $V_3 = 2V_1 = 14.96 \text{ L}$.

Leg 1→2 (isochoric): $W = 0$; $\Delta U = \frac{3}{2}nR\Delta T = \frac{3}{2}(0.60)(8.314)(300) = 2245 \text{ J}$; $Q = \Delta U = 2245 \text{ J}$.

Leg 2→3 (isothermal): $\Delta U = 0$; $W = nRT \ln \frac{V_3}{V_2} = (0.60)(8.314)(600) \ln 2 = 2075 \text{ J}$; $Q = W = 2075 \text{ J}$.

Leg 3→1 (isobaric): $W = p\Delta V = (2.0 \times 10^5)(7.48 - 14.96) \times 10^{-3} = -1497 \text{ J}$; $\Delta U = \frac{3}{2}nR\Delta T = \frac{3}{2}(0.60)(8.314)(-300) = -2245 \text{ J}$; $Q = \Delta U + W = -3741 \text{ J}$.

Leg	Process	Q (J)	W (J)	ΔU (J)
1→2	isochoric	2245	0	2245
2→3	isothermal	2075	2075	0
3→1	isobaric	-3741	-1497	-2245
cycle	total	578	578	0

Evaluate: $\sum \Delta U = 0$ and $\sum Q = \sum W = 578 \text{ J}$, as the first law requires for a cycle; the net work is positive, as predicted for a clockwise path. Heat is absorbed in legs 1→2 and 2→3 (4319 J) and rejected in 3→1 (3741 J).

A2. Adiabatic compression, then cooling

Air ($\gamma = 1.40$, treat as a diatomic ideal gas) at 1.0×10^5 Pa and 300 K fills a 2.0 L cylinder. It is compressed adiabatically to 0.50 L. It is then held at this volume while it cools back to 300 K. (a) Find the temperature and pressure after the compression and the work done on the gas. (b) Find the heat that leaves during the cooling and the final pressure. (c) Compare the total work with that of an isothermal compression between the same end states.

Solution. $n = \frac{p_1 V_1}{RT_1} = \frac{(1.0 \times 10^5)(2.0 \times 10^{-3})}{(8.314)(300)} = 0.0802$ mol. *Sketch:* a steep curve to the left and up, then a vertical line down to the isotherm through the start. *Predict:* the compression raises T above 300 K and the pressure by more than the factor 4.

(a) $\frac{V_1}{V_2} = 4$. $T_2 = T_1 4^{0.40} = 300 \times 1.741 = 522$ K; $p_2 = p_1 4^{1.40} = 1.0 \times 10^5 \times 6.96 = 6.96 \times 10^5$ Pa. $W = nC_V(T_1 - T_2) = (0.0802)(20.79)(300 - 522.3) = -371$ J (work done on the gas). (b) Constant volume: $W = 0$ and $Q = nC_V\Delta T = (0.0802)(20.79)(300 - 522.3) = -371$ J. Final pressure: $p_3 = p_2 T_3/T_2 = 6.96 \times 10^5 \times 300/522.3 = 4.0 \times 10^5$ Pa, which is $p_1 V_1/V_2$, as it must be for the same temperature. (c) Total work = -371 J (all in the adiabatic leg); total heat = -371 J; $\Delta U_{\text{total}} = 0$. A direct isothermal compression between the same states gives $W = nRT \ln(V_2/V_1) = (0.0802)(8.314)(300) \ln \frac{1}{4} = -277$ J. The difference (371 against 277 J) shows again that the work depends on the path, while $\Delta U = 0$ for both. *Check:* the adiabatic work $(p_1 V_1 - p_2 V_2)/(\gamma - 1) = (200 - 348)/0.40 = -370$ J. ✓

B. Practice

B1. Signs and systems

For each situation name the system, say whether it is open, closed or isolated, and give the sign of Q , W and ΔU for the system. (a) The gas in a sealed syringe is heated while the plunger is held fixed. (b) The air in a bicycle pump is compressed quickly and the barrel becomes hot. (c) A balloon floating in the sky expands as it rises (take the gas as the system and assume the walls are good insulators). (d) A cup of tea cools on a table.

B2. Work from areas

A gas goes from a (3.0 L, 2.0×10^5 Pa) to b (9.0 L, 0.50×10^5 Pa) along three paths: (1) a straight line; (2) a vertical drop to 0.50×10^5 Pa and then a horizontal line; (3) a horizontal line to 9.0 L and then a vertical drop. Sketch the paths and find the work done by the gas on each. Which is the greatest?

B3. An isothermal expansion

2.0 mol of an ideal gas at 300 K expands isothermally from 5.0 L to 20.0 L. Find the initial and final pressures, the work done by the gas and the heat that flows into it. Where does the energy come from?

B4. Working off a snack

A 65 kg student eats a snack worth 250 kcal and wants to use all of this energy by climbing stairs. Treat the student as the system with $\Delta U = 0$ and assume that all the energy is converted into the work of lifting. How high must the student climb? Comment on the assumption.

B5. Boiling and melting

(a) 10.0 g of water is boiled at 1.00 atm and becomes steam. Find Q , W and ΔU . (b) 100 g of ice at 0°C melts at 1.00 atm; its volume decreases from 109 cm^3 to 100 cm^3 . Find Q , W and ΔU , and explain the sign of W .

B6. Completing a ledger

A system takes a clockwise cycle $a \rightarrow b \rightarrow c \rightarrow a$. In $a \rightarrow b$, 500 J of heat is added and the system does 200 J of work. In $b \rightarrow c$ the volume is constant and the internal energy falls by 350 J. In $c \rightarrow a$, 100 J of heat is removed. Complete a table of Q , W and ΔU for each leg and for the cycle. What is the net work? Is it consistent with a clockwise cycle?

B7. Free expansion

2.0 mol of an ideal gas at 300 K is in an insulated rigid container, behind a partition, and the other half of the container is evacuated. The partition is removed and the gas fills the whole container, doubling its volume. (a) Find Q , W , ΔU and the final temperature. (b) The same end state is reached by a slow isothermal expansion against a piston. Find Q , W and ΔU for it. (c) What do the two cases show about Q , W and ΔU ?

B8. Heat at constant volume and at constant pressure

2.0 mol of oxygen (diatomic) is heated from 280 K to 380 K. Find Q , W and ΔU (a) at constant volume and (b) at constant pressure. (c) Repeat (b) for 2.0 mol of argon. (d) Why is ΔU different for the two gases while W is the same?

B9. A piston and a heater

A cylinder with a freely moving piston of area 0.020 m^2 holds 0.30 mol of nitrogen (diatomic) at a steady pressure of $1.20 \times 10^5\text{ Pa}$ and 300 K. A heater adds 800 J of heat. Find the temperature rise, the work done, the change in internal energy and the distance the piston moves.

B10. A rising bubble of air

Air at 25°C and 1.00 atm expands adiabatically to twice its volume ($\gamma = 1.40$). Find the final temperature and pressure. How do they compare with an isothermal expansion to the same volume?

B11. The pump, adiabatic limit

A pump holds 100 mL of air at 20°C and 1.00 atm. It is compressed to 25 mL so quickly that no heat escapes. Find the final temperature and pressure, and the work done on the gas ($\gamma = 1.40$).

B12. Argon expands adiabatically

2.0 L of argon at 5.0×10^5 Pa and 300 K expands adiabatically until its pressure is 1.0×10^5 Pa. Find the final volume and temperature, and the work done by the gas. How much more work would an isothermal expansion to the same final volume do?

B13. A four-leg cycle

0.050 mol of air (diatomic, $\gamma = 1.40$) is at 1.0×10^5 Pa and 300 K (state 1). It is (1→2) compressed adiabatically to $\frac{1}{8}$ of its volume; (2→3) heated at constant volume to 1500 K; (3→4) expanded adiabatically to the original volume; (4→1) cooled at constant volume to 300 K. Sketch the cycle, find p , V and T at each state, and Q , W and ΔU for each leg. Find the net work and the ratio of the net work to the heat absorbed.

B14. Challenge: the method is not given

1.0 mol of helium at 300 K and 2.0×10^5 Pa doubles its volume in three different ways: at constant pressure, at constant temperature, and adiabatically. For each, find the final temperature, W , ΔU and Q . Rank the three by the heat absorbed and explain the ranking in terms of the pV diagram.

Answers

- B1 (a) gas; closed; $Q > 0$, $W = 0$, $\Delta U > 0$. (b) air in the pump; closed; $W < 0$ (work done on the gas), $Q \approx 0$ for a fast stroke, $\Delta U > 0$ (the temperature rises). (c) gas in the balloon; closed; $W > 0$, $Q \approx 0$, $\Delta U < 0$ (it cools). (d) tea (a body of liquid, open to the air); open; $Q < 0$ (heat leaves), $W \approx 0$, $\Delta U < 0$
- B2 $W_1 = \frac{1}{2}(2.0 + 0.50) \times 10^5 \times 6.0 \times 10^{-3} = 750$ J; $W_2 = (0.50 \times 10^5)(6.0 \times 10^{-3}) = 300$ J; $W_3 = (2.0 \times 10^5)(6.0 \times 10^{-3}) = 1200$ J; path (3) is the greatest
- B3 $p_1 = nRT/V_1 = 9.98 \times 10^5$ Pa; $p_2 = 2.49 \times 10^5$ Pa; $W = nRT \ln 4 = 6.92 \times 10^3$ J; $\Delta U = 0$ so $Q = W = 6.92 \times 10^3$ J. The energy comes from the heat that enters from the surroundings (the thermal bath)
- B4 $Q = 250 \times 4186 = 1.05 \times 10^6$ J = mgh , so $h = 1.05 \times 10^6 / (65 \times 9.8) = 1.6 \times 10^3$ m (1.6 km). The assumption is unrealistic: muscles convert only about a quarter of the energy to mechanical work, so the useful climb is only about 0.4 km; the rest leaves the body as heat, and the bookkeeping still balances ($Q_{\text{in}} = W + Q_{\text{out}}$)
- B5 (a) $V_2 = 10 \times 1671 = 1.671 \times 10^4$ cm³, $W = p(V_2 - V_1) = (1.013 \times 10^5)(16.71 - 0.010) \times 10^{-3} = 1.69 \times 10^3$ J; $Q = mL_v = 22.56 \times 10^3$ J; $\Delta U = Q - W = 20.9 \times 10^3$ J. (b) $Q = mL_f = 33.4 \times 10^3$ J; $W = p\Delta V = (1.013 \times 10^5)(-9 \times 10^{-6}) = -0.91$ J (the atmosphere does a very small amount of work on the shrinking sample); $\Delta U = 33.4 \times 10^3$ J, essentially Q
- B6 a→b: $Q = 500$, $W = 200$, $\Delta U = 300$. b→c: $Q = -350$, $W = 0$, $\Delta U = -350$. c→a: $Q = -100$, $\Delta U = 0 - (300 - 350) = 50$, $W = Q - \Delta U = -150$. Cycle: $Q = 50$, $W = 50$, $\Delta U = 0$ (all in joules). Net work 50 J > 0: consistent with clockwise
- B7 (a) $Q = 0$, $W = 0$ (no force opposes the gas), $\Delta U = 0$, so $T = 300$ K (ideal gas). (b) $\Delta U = 0$; $W = nRT \ln 2 = 3.46 \times 10^3$ J; $Q = W = 3.46 \times 10^3$ J. (c) Same initial and final states, so the same ΔU (0), but different Q (0 against 3.46×10^3 J) and W : heat and work depend on the path

- B8 (a) $Q = \Delta U = nC_V\Delta T = 2.0 \times 20.79 \times 100 = 4.16 \times 10^3 \text{ J}$, $W = 0$. (b) $Q = nC_p\Delta T = 5.82 \times 10^3 \text{ J}$; $W = nR\Delta T = 1.66 \times 10^3 \text{ J}$; $\Delta U = 4.16 \times 10^3 \text{ J}$. (c) argon: $\Delta U = \frac{3}{2}nR\Delta T = 2.49 \times 10^3 \text{ J}$, $W = 1.66 \times 10^3 \text{ J}$, $Q = 4.16 \times 10^3 \text{ J}$. (d) $W = nR\Delta T$ depends only on the amount and ΔT ; $\Delta U = nC_V\Delta T$ depends on the degrees of freedom (the oxygen stores energy in rotation as well)
- B9 $\Delta T = Q/(nC_p) = 800/(0.30 \times 29.10) = 91.6 \text{ K}$; $W = nR\Delta T = 229 \text{ J}$; $\Delta U = Q - W = 571 \text{ J}$ (check: $nC_V\Delta T = 571 \text{ J}$); $\Delta V = W/p = 1.90 \times 10^{-3} \text{ m}^3$; distance $= \Delta V/A = 0.095 \text{ m}$ (9.5 cm)
- B10 $T_2 = 298.15 \times 2^{-0.40} = 226 \text{ K}$ (-47°C); $p_2 = 1.013 \times 10^5 \times 2^{-1.40} = 3.84 \times 10^4 \text{ Pa}$ (0.38 atm). Isothermal: 298 K and $5.07 \times 10^4 \text{ Pa}$ (0.50 atm); the adiabatic gas is colder and at lower pressure
- B11 $T_2 = 293.15 \times 4^{0.40} = 510 \text{ K}$ (237°C); $p_2 = 1.013 \times 10^5 \times 4^{1.40} = 7.05 \times 10^5 \text{ Pa}$ (6.96 atm); $n = 4.16 \times 10^{-3} \text{ mol}$; $W = nC_V(T_1 - T_2) = -18.8 \text{ J}$ (work done on the gas)
- B12 argon, $\gamma = \frac{5}{3}$: $V_2 = 2.0 \times 5^{3/5} = 5.25 \text{ L}$; $n = 0.401 \text{ mol}$; $T_2 = p_2V_2/nR = 158 \text{ K}$; $W = nC_V(T_1 - T_2) = 712 \text{ J}$. Isothermal (to 5.25 L): $W = nRT \ln(5.25/2.0) = 966 \text{ J}$, which is 254 J more
- B13 $V_1 = 1.247 \text{ L}$, $V_2 = V_3 = 0.1559 \text{ L}$, $V_4 = V_1$. $T_2 = 300 \times 8^{0.4} = 689 \text{ K}$, $p_2 = 18.4 \times 10^5 \text{ Pa}$; $T_3 = 1500 \text{ K}$, $p_3 = 40.0 \times 10^5 \text{ Pa}$; $T_4 = 1500 \times 8^{-0.4} = 653 \text{ K}$, $p_4 = 2.18 \times 10^5 \text{ Pa}$. 1 $\rightarrow$ 2: $Q = 0$, $W = -404 \text{ J}$, $\Delta U = 404 \text{ J}$. 2 $\rightarrow$ 3: $Q = 843 \text{ J}$, $W = 0$, $\Delta U = 843 \text{ J}$. 3 $\rightarrow$ 4: $Q = 0$, $W = 880 \text{ J}$, $\Delta U = -880 \text{ J}$. 4 $\rightarrow$ 1: $Q = -367 \text{ J}$, $W = 0$, $\Delta U = -367 \text{ J}$. Net $W = 476 \text{ J} = \text{net } Q$, $\sum \Delta U = 0$. $W_{\text{net}}/Q_{\text{in}} = 476/843 = 0.565 = 1 - 8^{-0.4}$
- B14 $V_1 = 12.5 \text{ L}$. Isobaric: $T_2 = 600 \text{ K}$, $W = nR\Delta T = 2.49 \text{ kJ}$, $\Delta U = 3.74 \text{ kJ}$, $Q = 6.24 \text{ kJ}$. Isothermal: $T_2 = 300 \text{ K}$, $W = nRT \ln 2 = 1.73 \text{ kJ}$, $\Delta U = 0$, $Q = 1.73 \text{ kJ}$. Adiabatic: $T_2 = 300 \times 2^{-2/3} = 189 \text{ K}$, $\Delta U = -1.38 \text{ kJ}$, $W = 1.38 \text{ kJ}$, $Q = 0$. Ranking by Q : isobaric > isothermal > adiabatic ($= 0$). On the pV diagram the isobaric path lies highest (the greatest work and also a rise in T), the adiabat lowest (the least work, with the gas cooling)